

SET B

JRU Form 30
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JOSÉ RIZAL UNIVERSITY

80 SHAW BOULEVARD
MANDALUYONG CITY

EXAMINATION BOOK

NAME

ANTHONY S. MIJARES

DATE

02 / 20 / 2016

INSTRUCTOR

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SUBJECT

ITC SOC

DO NOT TEAR ANY PAGE OUT OF THIS
EXAMINATION BOOK

PRELIM LONG QUIZ - ITC 506 (PREDICTIVE MODELLING)

INSTRUCTION:

CAREFULLY ANALYZE THE PROVIDED DATASET AND ANSWER EACH QUESTION IN AT LEAST 150 WORDS. YOUR RESPONSES WILL BE RATED AS FOLLOWS (1) CLEAR TECHNICAL REASONING AND UNDERSTANDING OF THE CONCEPTS (5) (2) INCLUDED LABELED DIAGRAMS TO SUPPORT AND ILLUSTRATE YOUR EXPLANATION (3) PE ANSWERS ARE WELL-STRUCTURED, PRECISE, AND WRITTEN IN YOUR OWN WORDS.

1. EXAMINE THE DATASET CAREFULLY AND DETERMINE WHICH ATTRIBUTE WOULD YOU SELECT AS THE TARGET VARIABLES (Y) TO CREATE THE MOST IMPACTFUL PREDICTIVE MODEL FOR THE SPECIFIED DOMAIN. PROVIDE A CONVINCING JUSTIFICATION FOR YOUR CHOICE

2. BASED ON THE TARGET VARIABLE YOU CHOSE IN QUESTION 1. WOULD YOU APPLY K-NEAREST NEIGHBOR (KNN) OR LINEAR REGRESSION TO BUILD YOUR MODEL? EXPLAIN YOUR CHOICE BY DISCUSSING THE RELATIONSHIP BETWEEN THE FEATURES AND THE TARGET (DRAW A SIMPLE DIAGRAM TO SUPPORT YOUR EXPLANATION)

3. DESCRIBE YOUR STRATEGY FOR THE
TRAINING AND TESTING OF THE MODEL
WHICH SPECIFIC EVALUATION METRICS
WOULD YOU USE, EXPLAIN YOUR
CHOICE

FIELD ID	DATE	CROP TYPE	B. AGRICULTURE	NITROGEN (N)	PHOSPHORUS (P)	POTASSIUM (K)	RAINFALL (mm)	YIELD (kg/ha)
AGR-201		MAIZE	42.5	90	42	43	200	12,500
AGR-202		WHEAT	38.2	75	38	35	150	9,200
AGR-203		SOYBEANS	55.0	25	60	48	310	11,800
AGR-204		RICE	78.4	120	50	65	1200	15,300

ANSWERS :

- 1.) considering, that all the elements, nitrogen, phosphorus, and potassium are something to ~~depend~~ is something that will make you wonder what the dataset is trying to pertain.
which can be the nutrients found in the soil where the crops are planted. on the other hand soil moisture and rainfall (mm) are something I found that light a bulb in my head that this dataset is describing, which I will choose is the most appropriate which is yield.
As this dataset shows which is the best crop to plant ^{or we} based on the soil condition and its ^{nutrients} ~~nutrients~~ how much will it yield.
Hence, the Yield is the best attribute to choose as the Target Variable (y).

2.) Since I did chose Yield as the target variable, the appropriate algorithm to chose is ~~k-Nearest Neighbor~~ to build ~~the~~ ~~model~~. As by ~~(choosing)~~ ~~(choosing)~~ identifying)

but yield is
numeric